Network 1: Feed-Forward Network

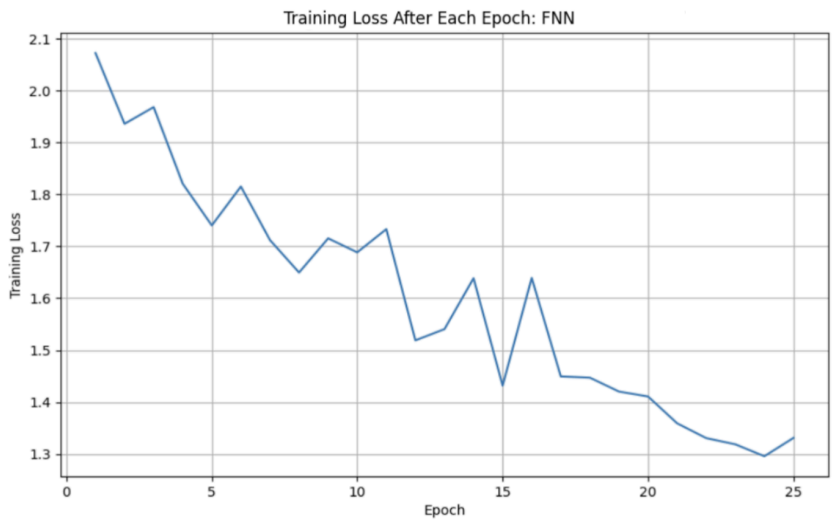
- Architecture:

  The first Network, Feed-Forward Network, is a fully connected neural network that has **no convolutional layers** and only contains **linear layers**.The input image is **flattened** into a one dimensional vector of **3072 features** by flattening its 32x32x3 pixels. The network consists of **three hidden layers of 512, 256, and 128 units**, followed by ReLU activation. There are ten units in the output layer, one for every CIFAR-10 class.

- Accuracy:

  Final Test Accuracy: 49.48%, **Epochs Trained:** 25

- Loss graph:



- Correct image and Incorrect image:

Network 2: Simple Convolutional Neural Network

- Architecture:

  The second network, Simple CNN, is a convolutional neural network with **two convolutional layers** for identifying spatial patterns in images. The first convolutional layer implements 32 filters with a 3x3 kernel, followed by ReLU activation and 2x2 max pooling to reduce the image size from 32x32 to 16x16. The second convolutional layer uses **64 filters** with a 3x3 kernel, followed by ReLU and 2x2 max pooling to reduce the image to **8x8**. After flattening, the network consists of two **fully connected hidden layers (512 and 128 units)** and an output layer with 10 units.

- Accuracy:

  Final Test Accuracy: 61.75%, **Epochs Trained:** 29

- Loss graph:

- Correct image and Incorrect image:

Network 3: Deep CNN

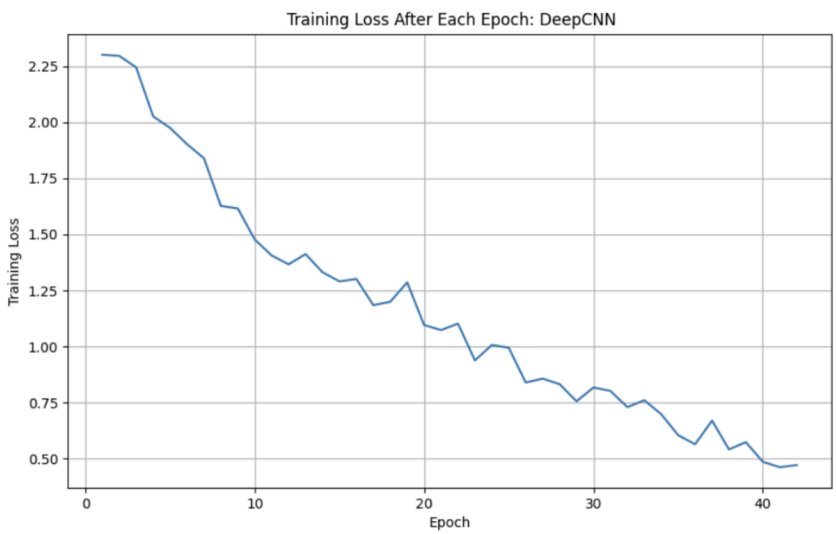
- Architecture:

  The third network, Deep CNN, is a deeper convolutional neural network with **three convolutional layers** and **dropout** regularization to avoid overfitting. With a 3x3 kernel, ReLU activation, and 2x2 max pooling, the three convolutional layers use 32, 64, and **128 filters**. As a result, the image shrinks from 32x32 to 16x16 to 8x8 to **4x4**. During training, the network randomly zeroes out neurons using **15% dropout** before and after the first fully connected layer. The fully connected layers consist of **two hidden layers (256 and 128 units)**, which are followed by a 10 unit output layer.

- Accuracy:

  Final Test Accuracy: 72.09%, **Epochs Trained:** 42

- Loss graph:



- Correct image and Incorrect image:

Comparison and analysis:

The **best performing network was Deep CNN** with 72.09% test accuracy. I believe that Deep CNN was able to get the best accuracy because its three convolutional layers have the ability to learn simple patterns first, like edges, then combine them into more complex shapes. Dropout regularization also helped avoid overfitting by making the network learn strong characteristics instead of memorizing the training data.

On the other hand, the **worst performing network was Feed-Forward Network** with 49.48% test accuracy. Since spatial detail is lost when an image is flattened into a one dimensional vector, I believe that the Feed-Forward Network scored the lowest accuracy. The network must handle each pixel independently as it is unable to recognize that nearby pixels form edges or patterns. As a result, learning useful features for image categorization becomes much more difficult.